CONTACT INFORMATION
1 (217) 305-2300
noejnava2@gmail.com
https://noejn2.github.io/

Citizenship: United States of America and México

POSITIONS

Seize — Institute of Consulting and Business Solutions 2025 - *present*
Economic Modeling Advisor
Riyadh, Saudi Arabia
Develop and deploy computable general equilibrium (CGE) and gravity models to assess trade agreements and policy reforms.
Build institutional modeling capacity, including AI-powered natural-language interfaces that let stakeholders query model results and run scenarios without writing code.
Pair data-driven and economics-driven analysis to inform policy decisions, and deliver capacity-building workshops for government staff.

Independent Consulting 2025
Economic Modeling Consultant
Chicago, IL

USDA Economic Research Service 2022 - 2025
Research Agricultural Economist
Market and Trade Economics Division,
Agricultural and Policy Models branch

EDUCATION

Ph.D., Agricultural and Applied Economics 2021
University of Illinois Urbana-Champaign, IL
Advisor: William Ridley
Dissertation: Three Essays on Agricultural and Consumer Economics,
Outstanding Doctoral Dissertation Award
Committee: William Ridley (chair), Sandy Dall'Erba, Yilan Xu, and Rafael Garduño-Rivera

B.A., Mathematical Economics 2016
California State University, San Bernardino, San Bernardino, CA
Advisor: Kazim Konyar
Summa Cum Laude

RESEARCH INTERESTS

Methods
Bayesian data analysis
General equilibrium models
Structural econometrics
Machine learning & AI for policy

Fields
Agricultural production
Consumption economics
Food systems

JOURNAL ARTICLES

1. *Land competition and poverty effects from the Mexico's proposal to ban Genetically Engineered corn.* 2024 **American Journal of Agricultural**

Economics 106(3): 1300-1325.
with Jayson Beckman, Angelica Williams, and Steven Zahniser.
doi:10.1111/ajae.12463

2. *Losing Ground: Measuring the Welfare Effects of Retail Food Price Inflation during the COVID-19 Pandemic on Mexican Household.* 2024 **Journal of Agriculture and Food Industrial Organization** 22 (2): 143-154.
with Benjamin D.K. Wood, and Rafael Garduño-Rivera.
doi:10.1515/jafio-2023-0041

3. *Estimation of the Value-Added/Intermediate Input Substitution Elasticities Consistent with the GTAP Data.* 2023. **Journal of Global Economic Analysis** 8(2): 134-158.
with Jayson Beckman, and Maros Ivanic.
doi:10.21642/JGEA.080204AF

4. *A model of the U.S. food system: What are the determinants of the state vulnerabilities to production shocks and supply chain disruptions?* 2023. **Agricultural Economics** 54(1): 95-109. [1] [2]
with William Ridley, and Sandy Dall'Erba.
doi:10.1111/agec.12750

5. *The impact of taxing sugary-sweetened beverages in México: A censored QUAI demand system approach.* 2022. **Journal of Agricultural and Applied Economics Association** 1(1): 18-32.
with Diansheng Dong.
doi:10.1002/jaa2.6

6. *U.S. interstate trade will mitigate the negative impact of climate change on crop profit.* 2021. **American Journal of Agricultural Economics** 103(5): 1720-1741.
with Sandy Dall'Erba, and Zhangliang Chen.
doi:10.1111/ajae.12204

7. *The impacts of agroforestry on agricultural productivity, ecosystem services, and human well-being in low- and middle-income countries: An evidence and gap map.* 2020. **Campbell Systematic Reviews** 16(1): e1066.
with Daniel C. Miller, Pablo J. Ordoñez, Sarah E. Brown, Samantha Forrest, Karl Hughes, and Kathy Baylis.
doi:10.1002/cl2.1066

8. *The conduits and barriers to reentry for formerly incarcerated individuals in San Bernardino.* 2018. **Journal of Prison Education and Reentry** 5(1): 2-17.
with Annika Anderson, and Patricia Cortez.
doi:10.25771/sdf0-1631

[1] Most Outstanding Second Year Research Paper at University of Illinois at Urbana-Champaign-ACE 2020.

[2] The Charles M. Tiebout Prize in Regional Science 2021.

Book Chapters

1. *Climate and Weather Impacts on Agricultural Productivity.* 2025. **Encyclopedia of Energy, Natural Resource, and Environmental Economics, 2e**.
with Eric Njuki, and Boris E Bravo-Ureta.

U.S. Department of Agriculture, Economics Research Service

1. *International Income and Price Elasticity Estimates: An Update* 2025 **Technical Bulletin** TB-1971.
with Yacob Abrehe Zereyesus, Tian Xia, Xianghong Li, and Lila Cardell
2. *Potential Budgetary Impacts of Climate Change on the Pasture, Rangeland, and Forage Insurance Plan* 2024 **Economic Brief** EB-38: 25pp.
with Dylan Turner, Katherine Baldwin, Jayson Beckman, Francis Tsiboe, and Kate Vaiknoras.
3. *ERS Research Models Future Effects of Climate Change on Corn and Soybean Yields, Production, and Exports* 2024 **Amber Waves** 1545-8741.
with Jayson Beckman.
ERS Amber Waves, March 04, 2024
4. *Estimating the market and trade implications from climate-Induced corn and soybean yield changes for the U.S.* 2023 **Economics Research Report** ERR-324: 31pp.
with Jayson Beckman, and Maros Ivanic.
5. *Warming temperatures in U.S. Corn Belt expected to continue into next decade.*
Charts of Note, December 5, 2023

Working Papers

1. *Zeros, Elasticities, and the Soda Tax: Estimator Choice in Censored Demand Systems.*
Solo paper. Under review at the **American Journal of Agricultural Economics**.
2. *Inflation and minimum-wage growth erode the nutritional gains from México's sugar-sweetened beverage tax.*
3. *Tracing U.S. county-to-county domestic trade flows in grains.*
with William Ridley, Sandy Dall'Erba, Jian Zou, Youning Chen, and Christa Court.
AAEA discussion paper
4. *The Impact of International Demand for Cereals on U.S. Crop Land Appreciation: A Model Implied Instrumental Variable Approach*
with William Ridley, and Stephen Morgan.
AAEA discussion paper
5. *Does trade openness affect global grain supply response?*
with Yacob Zereyesus, Jennifer Kee, Stephen Morgan, Nathan Hendricks, Michael Johnson, Lila Cardell, Jeremy Jelliffe, and Walter Acpangan.
AAEA discussion paper

6. *Heterogenous patterns of crop yield growth stagnation across U.S. counties in the next decade*
   Solo paper.

7. *Revisiting the impact of climate change on agriculture through spatially-varying and place-tailored Ricardian estimates.*
   with Sandy Dall'Erba, Chang Cai, and A. Stewart Fotheringham.
   CREATE discussion paper

Papers in Preparation

1. *Heterogeneity in Mexican beverage demand: A censored QUAIDS atlas across demographics and survey waves, 2012–2024.*
   Solo paper.

2. *Is agricultural insurance a safety net or a net of complacency amidst adaptation to extreme weather events?*
   with Francis Tsiboe, Dylan Turner, Katherine Baldwin, and Sharon Raszap.

Software & Public Tools

R & Julia packages

gravityGE — one-sector Armington-CES gravity model with general equilibrium effects (CRAN; Julia port *GravityGE.jl* on JuliaHub).
censoredAIDS — maximum-likelihood estimation of censored (Quadratic) Almost Ideal Demand Systems (CRAN).

Interactive dashboards

U.S. Domestic Agricultural Trade — interstate agricultural trade flows with gravity / general-equilibrium policy simulation.
Fertilizer Flows — visualization of global fertilizer trade flows.

AI tools

AI Paper Referee — large-language-model auditor that reviews applied-economics writing against published standards.

Fellowships, Grants and Scholarships

- The impact of climate change on agricultural production, productivity, and trade ($403,152) 2023
- Improving the capabilities of global trade models to analyze changes in land use/land cover and green house emissisions ($280,000) 2023
- A global agricultural supply-demand-climate-trade model calibrated to Baseline ($375,000) 2022
- Towards a data hub of global daily weather forecasts for the ERS (**PI**) ($30,000) 2022
- ERS' international supply forecasting ($30,000) 2022
- International food consumption patterns-implications for global food security and trade ($25,000) 2022
- The Charles M. Tiebout Prize in Regional Science ($1,000) 2021
- Cleo Fitzsimmons Award ($500) 2020
- Agricultural and Consumer Economics Department Grant ($400) 2019
- Marshall Scott Fellowship ($4,877.85) 2018
- Brockson Fellowship ($3,902.28) 2017

- The Proactive Recruitment in Introductory Science and Mathematics Travel Grant ($1,000) 2016
- The Proactive Recruitment in Introductory Science and Mathematics Research Grant ($2,000) 2015
- The John & June Kennedy Economics Scholarship ($1,500) 2015

Awards and Honors

- Outstanding Doctoral Dissertation Award, University of Illinois at Urbana-Champaign-ACE 2021
- The Charles M. Tiebout Prize in Regional Science 2021
- Farm foundation Agricultural Scholar 2021
- Most Outstanding Second Year Research Paper, University of Illinois at Urbana-Champaign-ACE 2020

Teaching and Mentorship Experience

Mentor at ERS Summer internship
Kristiina Ala-Kokko (Kansas State University) 2022

Teaching Assistant

ACE 240: Personal Financial Planning FA19
Instructor: Paul B Stoddard, Senior Lecturer
Lecture: 160 students
University of Illinois at Urbana-Champaign, IL
Undergraduate Credit: 4 Hours.

ACE 231: Food and Agribusiness Management SP18
Instructor:Paul B Stoddard, Senior Lecturer
Lecture: 160 students
University of Illinois at Urbana-Champaign, IL
Undergraduate Credit: 4 Hours.

ACE 341AMM: Issues in Agribusiness Markets and Management FA17
Instructor: Paul B Stoddard, Senior Lecturer
Lecture: 20 students
University of Illinois at Urbana-Champaign, IL
Undergraduate Credit: 3 Hours.

Research in the News

1. *Climate Impacts on Corn and Soybean Yields by Mid-2030s* in Progressive Farmer
2. *"Climate change is real;" Researchers warn extreme weather events could impact Illinois agriculture* in WCIA.com
3. *Trade Insurance* in Jayson Lusk, Food and Agricultural Economist, personal blog.

Research Assistantships

Interstate Trade as Mitigating Factor for the Effects of Climate Change on Illinois Agriculture

January 2020 - December 2021
How do the interstate trade flows of Illinois farm products respond to extreme weather events?
University of Illinois at Urbana-Champaign, IL
Supervisor: Prof. William Clifton Ridley

Work Experience

Intern
Department of Agriculture, U.S., Economic Research Service — Summer 2021
Market & Trade Economics Division
Agricultural Policy and Models Branch
Kansas City, MO

Intern
Department of Agriculture, U.S., National Agricultural Statistics Service — 2016 - 2017
Environmental, Economics and Demographics
Washington, DC

Intern
Department of Agriculture, U.S., Rural Development — 2015 - 2016
Rural Housing Service
Washington, DC

Selected Presentations

The Impact of International Demand for Agricultural Products on U.S. Cropland Appreciation: A Model-Implied Instrumental Variable Approach
Agricultural & Applied Economics Association (AAEA) 2024

The Impact of International Demand for Agricultural Products on U.S. Cropland Appreciation: A Model-Implied Instrumental Variable Approach
The International Agricultural Trade Research Consortium (IATRC) 2023

Heterogeneous Patterns of Crop Yield Growth Stagnation across U.S. Stagnation across U.S. Counties in the Next Decade
Agricultural & Applied Economics Association (AAEA) 2023
Track Session: Studying Climate Change Impacts from Different Methodological Perspectives

The probabilistic role of climate change on crop yield potential
FCSM Research & Policy Conference (FCSM) 2022

The impact of taxing sugar-sweetened beverages in México: A censored QUAI demand system
Agricultural & Applied Economics Association (AAEA) 2022
Track Session: Identification in Demand Estimations

A model of U.S. domestic crop trade to study state's food vulnerability to supply chain disruptions [3]
Agricultural & Applied Economics Association (AAEA) 2021
Western Regional Science Association (WRSA) 2021

[3] Previously titled *Who benefits from local agriculture?*

North American Regional Science Council (NARSC) 2020

Revisiting the impact of climate change on agriculture through spatially-varying and place-tailored Ricardian estimates

North American Regional Science Council (NARSC) 2021
Workshop RED in Mexico: Challenges of a New Era 2018

References

William Clifton Ridley, Ph.D.
Professor, Department of Agricultural and Consumer Economics
University of Illinois Urbana-Champaign, IL
wridley@illinois.edu

Sandy Dall'Erba, Ph.D.
Professor, Department of Agricultural and Consumer Economics
University of Illinois Urbana-Champaign, IL
dallerba@illinois.edu

Rafael Garduño-Rivera, Ph.D.
Professor, School of Economic and Administrative Sciences
Universidad Panamericana, México
rgardunor@up.edu.mx

Programming & Deployment

Julia, LaTeX, Markdown, Python, R, Stan, Stata, & SQL.
Docker, AWS, Azure, Shiny Server, Git & GitHub Actions (CI/CD).
LLM/AI tooling for natural-language model interfaces and research workflows.

Languages

English & Spanish.

Service and Professional Activities

Journal Refereeing

American Journal of Agricultural Economics; Economic Modelling; Discover Food;s Environmental Research Letters; Journal of Agricultural and Applied Economics Association; Nature Food; Organisation for Economic Co-Operation and Development (OECD); U.S. Global Change Research Program (USGCRP); World Development.

USDA - Economics Research Service

Member, ERS Modeling Team Committee
Member, ERS Brownbag Series Committee

Abstract Reviewer

AAEA Annual Meeting, 2025.
AAEA Annual Meeting, 2024.